

ROTHERA STATION, Antarctica

WMO: 890620

Lat: 67.566S

Lon: 68.130W

Elev: 108

StdP: 14.64

Time Zone: -3.00 (W03)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-10.9	-6.5	-17.1	2.2	-10.1	-12.5	2.8	-5.5	49.1	23.9	45.1	23.5	7.0	0	1.108

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	40.1	35.0	38.9	34.4	37.8	33.7	36.1	38.9	35.2	37.9	34.5	37.1	12.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
33.1	27.8	36.6	32.4	26.9	35.6	31.8	26.3	35.0	13.7	39.3	13.3	38.3	13.0	37.3	41.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 43.0	38.2	33.7	-13.9	43.6	7.5	1.5	-19.3	44.7	-23.7	45.5	-27.8	46.4	-33.3	47.4		
(5)			-14.2	38.7	6.8	1.5	-19.1	39.8	-23.1	40.6	-26.9	41.5	-31.9	42.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.2	34.6	33.4	30.7	26.8	25.2	20.1	16.3	14.9	18.1	23.0	27.5	32.3
	DBStd	9.59	2.22	2.74	3.89	5.09	6.09	8.32	9.90	10.72	11.21	7.72	4.92	3.01
	HDD50	9053	477	464	599	697	768	896	1044	1088	957	838	675	550
	HDD65	14528	942	884	1064	1147	1233	1346	1509	1553	1407	1303	1125	1015
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	14.4	10.3	12.7	14.0	14.5	16.3	15.8	14.9	15.2	17.3	16.9	14.0	11.0
Precipitation	PrecAvg	26.7	2.2	2.3	2.8	2.3	2.4	1.8	1.7	2.0	2.3	2.7	2.4	1.9
	PrecMax	35.6	3.9	3.1	4.3	3.7	4.1	3.8	3.0	3.8	3.1	5.1	4.0	4.2
	PrecMin	20.1	1.2	1.2	1.4	0.9	1.1	0.7	0.7	0.4	1.4	1.1	0.8	0.4
	PrecStd	4.3	0.7	0.6	0.7	0.7	0.9	0.7	0.6	0.7	0.6	1.0	1.0	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	40.9	40.3	37.2	38.1	35.4	35.3	34.4	35.6	36.4	38.0	40.6
		MCWB	36.2	35.6	36.3	33.9	34.6	32.5	32.0	31.7	32.8	33.2	33.3	34.0
	2%	DB	40.3	39.3	38.1	35.5	35.6	33.8	33.0	32.7	33.9	34.7	35.7	38.7
		MCWB	35.1	35.0	34.6	32.6	32.7	31.3	30.5	30.8	31.7	32.3	32.6	33.5
	5%	DB	39.2	38.0	36.7	34.2	34.0	32.4	31.6	31.4	32.9	33.6	34.3	37.5
		MCWB	34.4	34.2	33.6	31.8	31.7	30.3	29.6	29.8	31.0	31.5	31.4	32.8
	10%	DB	38.0	37.0	35.6	33.1	32.8	30.9	29.9	29.4	31.7	32.5	33.2	36.4
		MCWB	33.7	33.5	32.9	31.0	31.0	29.1	28.2	27.7	30.1	30.7	30.5	32.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	37.1	36.9	34.4	35.3	33.0	32.4	32.3	33.2	33.8	34.4	35.9
		MCDB	40.8	39.6	39.7	36.3	38.0	34.9	34.4	33.7	34.7	35.5	36.6	38.8
	2%	WB	36.0	35.9	35.1	33.2	33.2	31.9	31.2	31.3	32.2	32.8	33.1	34.6
		MCDB	39.1	38.3	37.6	34.9	35.0	33.4	32.8	32.6	33.5	34.3	35.1	37.5
	5%	WB	35.2	35.0	34.1	32.3	32.1	30.7	29.9	29.8	31.3	32.0	32.2	33.8
		MCDB	38.1	37.4	36.2	34.0	33.6	32.4	31.4	31.4	32.7	33.4	34.0	36.5
	10%	WB	34.4	34.2	33.2	31.3	31.2	29.1	28.2	27.7	30.1	31.0	31.2	32.9
		MCDB	37.2	36.6	35.3	33.0	32.7	30.8	29.8	29.3	31.7	32.5	33.1	35.7
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.4	4.9	5.1	5.9	7.4	10.0	10.1	9.5	7.4	6.1	5.4
		MCDBR	6.7	6.4	5.6	6.1	6.4	7.1	10.3	9.2	8.5	7.1	6.8	6.6
		MCWBR	4.6	4.5	4.3	5.4	5.5	6.7	9.6	8.5	8.0	6.3	5.6	4.6
	5% WB	MCDBR	6.1	6.0	5.4	5.8	6.4	6.9	10.6	9.1	8.7	6.7	6.5	6.1
		MCWBR	4.4	4.4	4.4	5.5	5.6	6.7	10.3	8.7	8.5	6.2	5.6	4.5
Clear-Sky Solar Irradiance	taub	0.278	0.273	0.255	0.217	N/A	N/A	0.149	0.208	0.245	0.262	0.268	0.277	
	taud	2.373	2.395	2.387	2.229	N/A	N/A	1.900	2.095	2.229	2.228	2.262	2.296	
	Ebn at Noon	310	292	264	197	N/A	0	52	189	259	292	311	314	
	Edh at Noon	33	28	21	12	N/A	0	3	14	24	32	36	37	
All-Sky Solar Radiation	RadAvg	1992	1319	658	197	28	0	10	129	534	1320	2046	2405	
	RadStd	95	89	45	16	2	0	1	11	32	51	97	144	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page